

Proactive Memory for Event-Driven LLM Agents

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Abstract

Proactive LLM agents anticipate service opportunities from ongoing observations and accumulated history. A central challenge is determining which historical observations remain applicable to the current event and can support a timely service decision. We propose Proactive Evidence Memory (ProEviMem), combining bounded recent events, source-checked daily trajectories, and replay-assessed temporal patterns. Daily validation checks extracted facts against source events, while historical replay assesses recurring patterns before they are matched to current events. We also construct ProactiveStream, a longitudinal benchmark covering three application scopes, where agents process isolated event streams chronologically under episode-aware opportunity labels. ProEviMem improves Intervention Score by 4.3–11.5 percentage points over the strongest baseline in each scope. Ablation studies support the contributions of hierarchical memory organization and validation to service-opportunity detection.

1 Introduction

Proactive LLM agents anticipate useful assistance as a user’s situation evolves, taking initiative before an explicit request. Recent systems study when to propose a task, detect emerging needs, and use contextual or historical information to guide that judgment (Lu et al., 2025; Yang et al., 2025; Xie et al., 2026; Li et al., 2026). In activity support, ambient living, and repository maintenance, this requires interpreting a continuing stream of observations and deciding whether a service opportunity warrants attention now.

Historical context is central to this decision. Figure 1 illustrates how earlier evening walks can inform a service judgment when current activity is low and the user is available. Daily trajectories and temporal patterns retain this context beyond a

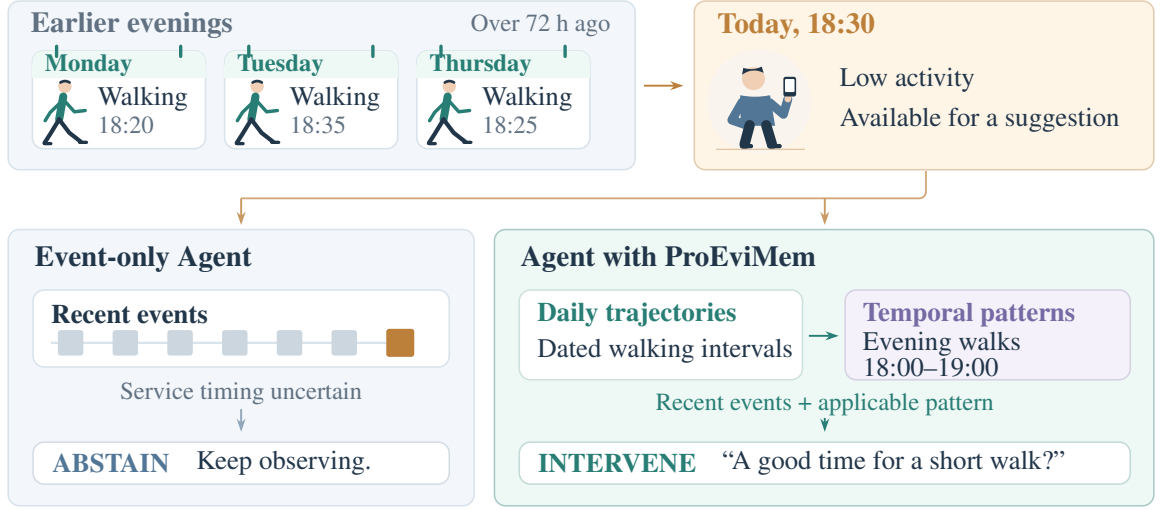
recent-event window. The agent must connect that history to a concrete, timely service.

Agent memory systems retain observations, consolidate experience, and retrieve related information (Park et al., 2023; Packer et al., 2023; Chhikara et al., 2025; Xu et al., 2025). To turn this history into decision context, proactive agents must assess whether an abstraction reflects its source events, whether a proposed temporal pattern has recurring support, and whether that pattern applies now. Our central question is *how to turn longitudinal history into reliable, currently applicable context for proactive service decisions*.

We propose Proactive Evidence Memory (ProEviMem; Figure 2), a hierarchical memory framework that organizes event histories into source-checked daily trajectories and replay-assessed temporal patterns for proactive service decisions. Bounded Event Memory preserves recent observations, while Daily Trajectory Memory retains source-checked abstractions beyond the recent window. Temporal Pattern Memory consolidates recurring conditions from these daily trajectories and assesses their historical support through replay. At decision time, the agent combines recent events and daily trajectories with currently applicable patterns. The decision LLM assesses whether current observations or historical context support a concrete, timely, low-risk service; episode control tracks whether the proposed opportunity has already been covered.

We construct *ProactiveStream* to study these decisions over 4,539 events from 17 isolated streams. Its three scopes preserve causal history and define episode-aware reference opportunities, allowing evaluation of memory formation and repeated service decisions within the same streams. ProEviMem improves Intervention Score (IS) by 4.3–11.5 percentage points over the strongest comparator in each scope, among five external baselines and a recursive-summary control. Hierarchy and valida-

From past activity to timely assistance



Illustrative example

Figure 1: Illustrative activity-support scenario. Both agents receive the same current event. ProEviMem supplements recent events with daily trajectories and temporal patterns built from earlier walking observations, informing the assessment of a timely activity suggestion. The two responses depict possible decisions in this conceptual example.

tion ablations examine how historical organization and support checks contribute to these gains.

Our contributions are:

- **Method.** ProEviMem combines source-checked daily abstraction, replay-based pattern assessment, and selection of currently applicable patterns to supply historical context for proactive service decisions.
- **Benchmark.** ProactiveStream supports longitudinal evaluation of memory formation and repeated service decisions, preserving causal history, entity isolation, and episode-aware reference opportunities across three scopes.
- **Empirical findings.** ProEviMem achieves higher IS than the evaluated baselines and summary control in all three scopes; targeted ablations support the contributions of hierarchical organization and validation.

2 Problem Formulation

Let u denote an independently evaluated entity, such as a participant, household, or repository, with a time-ordered public event stream

$$E^u = (e_1^u, \dots, e_{T_u}^u), \quad e_t^u = (\tau_t^u, x_t^u), \quad (1)$$

where T_u is the number of events for entity u , τ_t^u is the event time, and x_t^u contains the scope-specific

public fields observable at that time. We omit the entity superscript u when it is unambiguous. Events follow a fixed chronological order with deterministic tie-breaking. Before processing e_t^u , an agent has a causal state Z_t^u constructed only from events and finalized outputs preceding e_t^u in this order. It produces

$$(\hat{y}_t, \hat{m}_t) = \pi(e_t^u, Z_t^u), \quad \hat{y}_t \in \{0, 1\}, \quad (2)$$

where $\hat{y}_t = 1$ denotes INTERVENE, $\hat{y}_t = 0$ denotes ABSTAIN, and $\hat{m}_t$ is an optional evidence-grounded service message. Here π denotes a general inference policy over the current event and causal state; $\hat{m}_t$ may be empty under ABSTAIN. After finalizing the decision and optional message, the agent updates its state through an update function U :

$$Z_{t+1}^u = U(Z_t^u, e_t^u, \hat{y}_t, \hat{m}_t). \quad (3)$$

Each entity maintains isolated state, and π accesses only the current public event and causally available same-entity information.

The objective is to identify a timely, actionable *service opportunity* supported by the current event or prior same-entity observations. Let $y_t \in \{0, 1\}$ denote the frozen, reviewed reference decision for e_t under the annotation protocol in Section 4.2; all evaluated policies share these fixed references.

Labels and annotation metadata remain evaluator-private. Events with either reference label share the same scope-specific public input schema.

3 Proactive Evidence Memory (ProEviMem)

ProEviMem uses three complementary memory layers (Figure 2): Event Memory for recent public events, Daily Trajectory Memory for source-checked daily abstractions, and Temporal Pattern Memory for replay-assessed recurring patterns. Construction checks source fidelity and historical support, retrieval selects currently applicable patterns, and the decision LLM assesses present service value.

3.1 Memory Construction and Update

Bounded Event Memory. For event e_t , Event Memory retains prior events in their original public representation, intersecting a temporal horizon Δ with capacity K :

$$S_t = \text{Tail}_{K-1}((e_i)_{i < t, 0 \leq \tau_t - \tau_i \leq \Delta}). \quad (4)$$

Here Tail_{K-1} selects the most recent $K-1$ events in the filtered sequence. We use $\Delta = 72$ hours and $K = 100$: at most 99 prior events accompany the separately supplied current event. An independent daily buffer retains all committed events of the open local day, preserving complete input for daily construction.

Fact-validated Daily Trajectory Memory. After a local day closes, a construction LLM summarizes its buffered events as trajectory segments with intervals, typed facts, and event citations. Let $F(\cdot)$ denote normalized checkable facts obtained from typed fields. For segment d and its cited events $\mathcal{E}(d)$, source consistency requires

$$F(d) \subseteq \bigcup_{e \in \mathcal{E}(d)} F(e), \quad \mathcal{E}(d) \subseteq E_{\text{completed day}}^u. \quad (5)$$

Here $E_{\text{completed day}}^u$ contains the same entity’s events from that completed day. The validator checks entity, interval, citations, and canonicalized scalar values at compatible field paths. Acceptance requires source-supported typed facts and resolved citations; free text provides accompanying descriptions. Accepted daily trajectories form W_t , retaining historical context beyond Event Memory. Appendix E details field handling and buffering.

Replay-assessed Temporal Pattern Memory.

Weekly consolidation groups seven completed daily records per entity in time order, with each record assigned to one cycle. A cycle may span more than seven calendar days. An LLM proposes a trigger g , expected pattern p , time window, optional weekday restrictions, and source daily identifiers. Replay checks these conditions against validated daily facts: an observable triggered day is fulfilled when p occurs at or after the trigger within the window and violated otherwise. Missing observations are censored, inapplicable days are excluded, and each date contributes at most once. Let n_f count fulfilled (supporting) days and n_v count violated days:

$$\text{conf}_{\text{replay}} = \frac{n_f}{n_f + n_v}, \quad \text{lift} = \frac{\text{conf}_{\text{replay}}}{\text{base}(p)}. \quad (6)$$

The base rate is the frequency of p over observable days under the same window/weekday restrictions across both triggered and other observable days. Zero denominators yield zero confidence or lift, respectively.

A well-formed proposal becomes *Active* when $n_f \geq 3$, confidence ≥ 0.5 , and lift ≥ 1.0 ; otherwise it remains decision-invisible *Candidate*. Malformed proposals are rejected. Consolidation resets the online daily view, retaining records for audit. Later observations update pattern confidence and lifecycle: repeated violations can lower retrieval priority or suspend eligibility. Appendix E.3 specifies these transitions. Historical support characterizes observed recurrence; future applicability and present service value require separate assessment.

3.2 Memory Retrieval and Selection

Event Memory supplies the bounded sequence and W_t supplies all completed daily trajectories in the current cycle. Pattern retrieval additionally checks entity-local trigger context, weekdays, and temporal applicability with a 30-minute lead/grace interval. Matching considers current and previously recorded causal trigger occurrences. An occurrence is fulfilled when its expected behavior has been observed; censored occurrences have insufficient observations for assessment. Selection excludes both states. For pattern store L_t^u , lifecycle eligibility Eligible_t , trigger/time matching Match_t , and

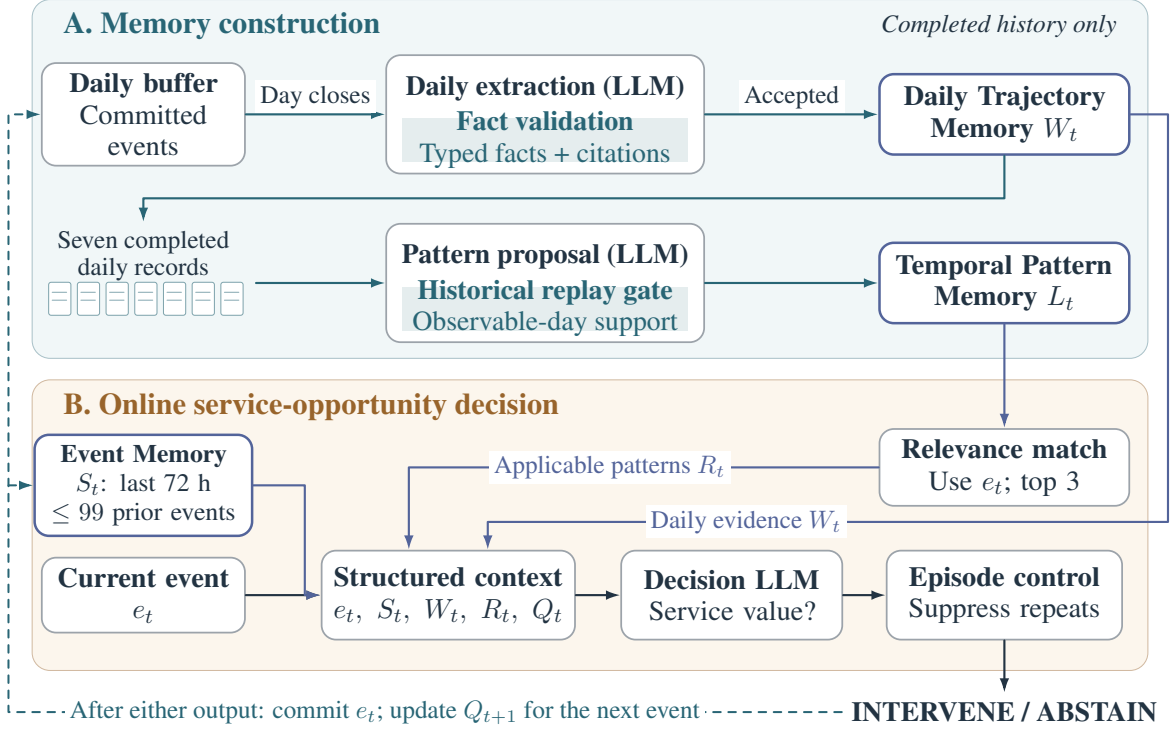


Figure 2: ProEviMem separates memory construction (A) from online decision-making (B). Daily extraction is checked against cited events; temporal proposals are assessed by replay over completed daily records. Only lifecycle-eligible, currently relevant patterns enter the structured context. The decision LLM assesses service value, and the shared controller filters proposed interventions to suppress repetition. Dashed feedback commits the current event after the output is finalized, whether INTERVENE or ABSTAIN; Q_t describes observable episode state.

occurrence state s_t ,

$$R_t = \text{Top3} \{ \ell \in L_t^u : \text{Eligible}_t(\ell) \wedge \text{Match}_t(\ell), s_t(\ell) \notin \{\text{fulfilled, censored}\} \}. \quad (7)$$

The top-three selection ranks eligible patterns by occurrence state, lifecycle, confidence, and temporal distance (Appendix E.4). All matching uses causally available observations to assess present applicability.

3.3 Proactive Decision

The decision context combines the current event, recent events, daily trajectories, selected patterns, and observable episode state:

$$C_t = (e_t, S_t, W_t, R_t, Q_t), \quad (8)$$

where Q_t describes the service opportunity’s observable state; the controller separately records previously allowed interventions to track service coverage. The LLM produces $(\tilde{y}_t, \tilde{m}_t) = \pi_\theta(C_t)$, proposing intervention when the current event *or* historical context supports concrete, timely, feasible, low-risk assistance, and abstention otherwise.

Service value governs intervention across both routine and deviating situations.

Within each scope, all methods apply the same Episode Controller policy to their own context and intervention history. The controller compares proposed services with previously allowed interventions and converts duplicates to ABSTAIN. Scope-specific changes in opportunity state, service identity, contextual patterns, or local day permit renewed assessment; each intervention originates from an LLM proposal. The controller finalizes the proposal as $(\hat{y}_t, \hat{m}_t)$, the output defined in Section 2. Appendix E.5 specifies matching fields, phase transitions, and re-entry rules.

Inference prepares prior memory, stages e_t , selects relevant context, obtains and finalizes the decision, and then commits e_t . Maintenance incorporating e_t can affect only later decisions. Reference labels remain evaluator-private throughout construction and retrieval.

4 ProactiveStream Benchmark

ProactiveStream evaluates service-opportunity detection in longitudinal event streams, where agents

assess incoming observations using their own accumulated history. It comprises 4,539 events across 17 isolated streams in three service scopes, with 729 reviewed reference interventions (Table 1).

4.1 Scopes and Public Events

Activity Support (AS) uses participant activity and availability contexts for optional activity assistance.

Ambient Living (AL) uses time-bounded home-sensor observations for gentle, low-risk check-ins. Each sensor segment becomes available at interval closure.

Repository Assistance (RA) uses repository activity at GitHub’s native event granularity for low-risk triage, review, and maintenance, with one decision per event.

Each participant, home, or repository defines an isolated chronological stream, allowing earlier observations to inform later service opportunities. Public inputs contain contemporaneously available observations, using the same scope-specific schema for both label classes. Agents construct their own memory from these events; annotation and scoring metadata remain evaluator-private. Appendix F details source datasets, fields, and preprocessing.

4.2 Reference Labels and Causal Replay

Annotators assess e_t and its complete prior same-stream history $\mathcal{H}_{<t}$. Let A indicate a concrete, timely, feasible, low-risk opportunity supported by e_t , B an opportunity at e_t supported by historical observations, and N_t^{ref} reference-episode novelty:

$$y_t = 1 \iff (A(e_t) = 1 \vee B(e_t, \mathcal{H}_{<t}) = 1) \wedge N_t^{\text{ref}} = 1. \quad (9)$$

The reference protocol labels the first reliable actionable window INTERVENE and covered windows ABSTAIN, allowing renewed opportunities when support strengthens, the service changes, or an episode reopens. Reference novelty follows the reviewed opportunity sequence independently of model outputs and is shared across methods. The event-level labels therefore assess timely identification of reviewed service windows.

LLM-assisted screening prioritizes candidate opportunities; three team members then independently label every event while blinded to evaluated-system outputs, followed by cross-review and adjudication. Appendix D reports a separate-team review of three contiguous 150-event segments

with their preceding histories. Main comparisons use protocol-fixed evaluation on the jointly developed benchmark, with labels, prompts, memory bounds, and validation settings frozen before the corresponding runs.

Each stream is replayed chronologically from empty method state, with access to the current event and strictly prior same-stream context. Each decision is finalized before the current event is committed to memory; reference labels are joined after inference for scoring. Scoring includes the initial events and continues as history accumulates.

5 Experimental Evaluation

5.1 Experimental Settings

We examine overall effectiveness (RQ1), memory-level contributions (RQ2), validation mechanisms (RQ3), and temporal availability and decision behavior (RQ4).

Baselines. We evaluate five external methods:

- **ProactiveAgent** uses the released evaluation prompt and proposal/null contract with the study backbone, retaining causal event-decision history capped at 1,000 events (Lu et al., 2025; THUNLP, 2026).
- **ContextAgent-ICL** adapts ContextAgent’s in-context proactive-service judgment to public event contexts using the study backbone (Yang et al., 2025).
- **Mem0** uses open-source extraction, consolidation, and same-entity retrieval (Chhikara et al., 2025; Mem0, 2026).
- **A-MEM** constructs linked memory notes, updates their associations as events arrive, and retrieves relevant notes (Xu et al., 2025).
- **Graphiti** stores event-derived relations in a temporal graph and retrieves same-stream facts (Zep, 2026).

Summary-memory control. **Cross-week Summary Memory (CSM)** recursively consolidates daily and cross-week summaries, providing a plain-summary control for retaining history beyond the recent-event window. Its results appear beside the external baselines in Table 2.

Table 1: Composition and provenance of ProactiveStream. AS: Activity Support; AL: Ambient Living; RA: Repository Assistance. Each stream is an independent memory and replay unit. Reference interventions denote team-reviewed service opportunities.

Scope	Primary source	Entity	Streams	Input events	Temporal coverage	Reference interventions
AS	HeartSteps	participant	9	1,456	367 observed participant-days	365
AL	CASAS	home	4	1,624	14 consecutive days/home	190
RA	GH Archive	repository	4	1,459	longitudinal repository histories	174
Total	three scopes	isolated stream	17	4,539	three longitudinal settings	729

Memory variants. **Event-only** supplies the current event and bounded Event Memory; **Event+Daily** additionally supplies Daily Trajectory Memory. **Flat Retrieval Memory** augments Event-only with three similar older same-entity events on AS/AL, using fixed dense embeddings and cosine similarity. All methods share the task semantics and final Episode Controller, with method-specific memory interfaces. Event-only, Event+Daily, and ProEviMem also share their decision prompt, schema, and Event Memory bound. CSM, A-MEM, and Graphiti augment that same event bound with their respective representations. Appendix G records interface and retrieval settings.

Model settings. The primary model is gpt-5.6-luna, with temperature zero and thinking disabled; ProEviMem uses the official OpenAI API. Terra/Sol comparisons replace both decision and memory-construction models (Appendix A).

Metrics. We report results separately for AS, AL, and RA using final decisions after episode control. INTERVENE includes immediate and scheduled service; ABSTAIN denotes silence at the current event. **Precision** measures the share of interventions matching reference opportunities. **Opportunity Detection Rate (ODR)** measures the fraction of reference-positive events identified for intervention. **Intervention Score (IS)** is the harmonic mean of precision and ODR. ODR and IS are task-oriented names for standard recall and F1, respectively. Appendix G provides their formulas. **False Positive Rate (FPR)**, also termed False Intervention Rate (FIR), is $FP/(FP + TN)$. **Balanced Accuracy (BA)** averages ODR and specificity. Higher precision, ODR, IS, and BA and lower FPR are preferred. **Intervention Rate (IR)** is the fraction of events receiving an intervention

and describes decision frequency.

5.2 Main Results

ProEviMem achieves the highest IS and BA in all three scopes (Table 2, RQ1). Its IS gains over the strongest comparator in each scope range from 4.3 to 11.5 percentage points. The comparisons below examine how opportunity recovery and intervention selectivity contribute to this service-opportunity detection performance.

Comparisons with external methods. The IS advantage spans proactive prompting, extracted memories, linked notes, and temporal-graph retrieval. On AS, ProEviMem attains the lowest FPR among all comparators; on AL, it combines the highest precision and ODR. Relative to A-MEM, it recovers additional reference opportunities while reducing false interventions in every scope. These results support selecting historical information that is applicable to the current service need. The comparisons assess complete memory interfaces; the controlled ablations below examine individual design choices.

Cross-week summary control. CSM improves on Event-only in every scope, showing the value of retaining compressed history beyond the recent-event window. ProEviMem further improves IS and precision across all three scopes. On RA, ProEviMem retains 89.7% ODR while producing 179 fewer false interventions than CSM, with nine fewer true positives. This trade-off yields an 11.5-point IS gain, supporting more selective use of retained history. Section 5.3 tests how source validation and replay gating contribute to this selectivity.

Table 2: Main results (% , Luna). IR: Intervention Rate; ODR: Opportunity Detection Rate; IS: Intervention Score; FPR: False Positive Rate; BA: Balanced Accuracy. Bold marks the best directional score. IR describes decision frequency. CSM is the cross-week summary control.

Scope	Method	IR	Precision $\uparrow$	ODR $\uparrow$	IS $\uparrow$	FPR $\downarrow$	BA $\uparrow$
AS	ProactiveAgent	69.0	30.0	82.5	43.9	64.5	59.0
	ContextAgent-ICL	54.3	31.0	67.1	42.4	50.0	58.6
	Mem0	85.2	28.8	97.8	44.5	81.0	58.4
	A-MEM	60.5	30.6	74.0	43.3	56.0	59.0
	Graphiti	55.1	29.3	64.4	40.3	52.0	56.2
	CSM	58.9	30.8	72.3	43.2	54.4	59.0
	ProEviMem	53.8	35.8	76.7	48.8	46.1	65.3
AL	ProactiveAgent	9.2	34.2	26.8	30.1	6.8	60.0
	ContextAgent-ICL	16.1	35.9	49.5	41.6	11.7	68.9
	Mem0	8.5	23.9	17.4	20.1	7.3	55.0
	A-MEM	19.0	34.6	56.3	42.9	14.1	71.1
	Graphiti	17.9	33.4	51.1	40.4	13.5	68.8
	CSM	17.1	43.7	63.7	51.8	10.9	76.4
	ProEviMem	17.7	48.6	73.7	58.6	10.3	81.7
RA	ProactiveAgent	53.1	17.5	78.2	28.7	49.7	64.2
	ContextAgent-ICL	32.0	22.5	60.3	32.8	28.2	66.1
	Mem0	7.7	39.8	25.9	31.4	5.3	60.3
	A-MEM	30.4	30.4	77.6	43.7	24.0	76.8
	Graphiti	28.1	26.8	63.2	37.7	23.3	69.9
	CSM	39.5	28.6	94.8	43.9	32.1	81.4
	ProEviMem	26.7	40.1	89.7	55.4	18.1	85.8

5.3 Ablation Study

Memory levels. Figure 3 compares Event-only, Event+Daily, and ProEviMem with the same Luna model, Event Memory bound, decision prompt, schema, and controller. Daily trajectories improve opportunity recovery, with the largest gain on RA. Adding Temporal Pattern Memory increases IS by 9.9, 13.5, and 7.0 points on AS, AL, and RA, respectively. The pattern layer gains true positives and reduces false positives in each scope. These complementary effects support organizing longer history into currently applicable patterns (RQ2).

The daily layer leaves false-positive counts unchanged on AS/AL and adds two on RA, explaining its nearly flat FPR. It primarily expands recoverable context. The pattern layer adds historical support and applicability selection, improving both opportunity recovery and selectivity. Net count changes summarize aggregate differences between variants.

Validation mechanisms. Table 3 tests daily fact validation and replay gating under the same model, prompt, event bound, context capacity, and episode control. *w/o Replay Gate* admits well-formed, source-linked, online-reconstructable Candidates by bypassing Equation 6’s activation thresholds; current-applicability matching remains enabled. *w/o Daily Validation* passes schema-valid, same-

entity daily extractions directly to storage, retaining weekly construction and replay.

Both checks improve precision and IS across scopes (RQ3). Replay gating produces the larger IS contribution and improves ODR in every scope. Under a fixed context capacity, replay-based admission may improve ODR by prioritizing historically supported patterns for the decision model. On RA, ProEviMem produces 43 fewer false interventions than *w/o Daily Validation* while retaining the same true-positive count. This comparison supports the contribution of source-checked daily extractions to downstream selectivity.

Episode control. The separate *Before suppression* diagnostic measures the controller’s action on recorded RA proposals. Suppression removes 66 false-positive and 17 true-positive proposals, increasing precision and IS while lowering event-level ODR. This operating-point change concerns duplicate-opportunity control; the sequential memory states and later decisions remain fixed in this diagnostic.

5.4 Temporal Memory and Decision Behavior

When longer history becomes usable. The AS/AL audit records 423 daily trajectories and 42 Active temporal patterns. Median times to first

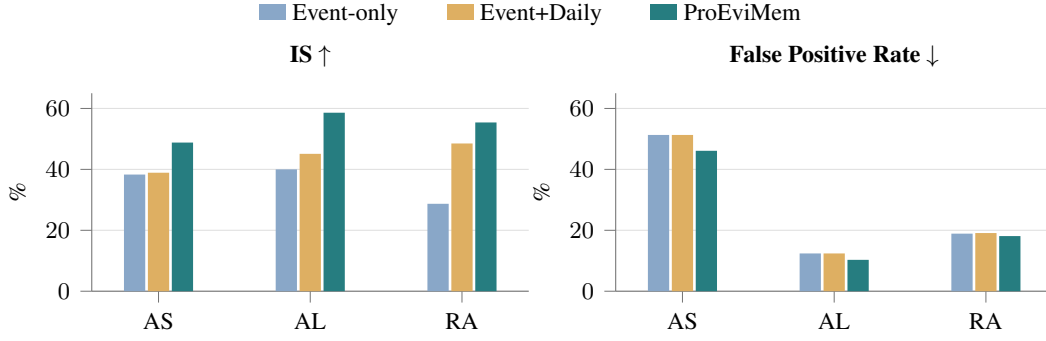


Figure 3: Memory-level ablation (%). Daily trajectories increase IS while leaving FPR nearly unchanged; temporal patterns further increase IS and reduce FPR in every scope. Model, prompt, event bound, and controller are shared. Gains are computed before rounding. Appendix H reports exact scores and net TP changes.

Table 3: Validation ablations and recorded controller diagnostic (%). Precision and IS are higher-is-better. Validation variants remove their named check. Before suppression scores recorded RA proposals while preserving the original run’s later states and decisions. Bold marks the best score; Appendix H adds IR and ODR.

Method	AS		AL		RA	
	Precision	IS	Precision	IS	Precision	IS
w/o Replay Gate	31.1	42.7	43.6	54.4	33.8	48.1
w/o Daily Validation	34.4	47.5	45.9	56.7	36.1	51.5
Before suppression	—	—	—	—	36.7	53.6
ProEviMem	35.8	48.8	48.6	58.6	40.1	55.4

daily trajectory, Active pattern, and relevant pattern are 0.80, 6.79, and 10.46 days, calculated over 13, 11, and eight entities reaching the respective stages. These stage-specific summaries describe construction, admission, and first online matching across distinct eligible-entity sets. They motivate preserving longitudinal history for both memory construction and subsequent use (RQ4). Appendix B gives the full counts and eligible-entity denominators.

From historical pattern to current judgment.

A recorded AS case links five days of STILL observations, completed daily trajectories, and an activated afternoon pattern. At a later matched event, the participant has 778 recorded steps in the preceding hour; the decision model returns ABSTAIN, which the controller preserves. The current activity provides context for interpreting the earlier pattern. This case illustrates how applicable memory informs service-value judgment while the final decision responds to the current observation. Appendix C traces the timestamps and source links.

Intervention frequency and selectivity. For reference-positive prevalence ρ , the identity $IR = \rho ODR + (1 - \rho) FPR$ separates opportunity recovery from false intervention. AL and RA have similar prevalences, yet ProactiveAgent intervenes

much more frequently on RA. Together with the main comparisons, this shows that intervention frequency reflects both the service setting and the method’s selectivity. The hierarchy improves this balance by conditioning decisions on supported, currently applicable history.

Additional comparisons. Flat Retrieval Memory improves on Event-only in AS/AL, while ProEviMem achieves higher precision and ODR than this retrieval control. Replacing both decision and construction LLMs from Luna to Terra to Sol also improves IS and FPR in every scope (Appendix A).

Reference-label and episode diagnostics. A separate three-reviewer team assesses 150 consecutive targets per scope with complete prior history and model outputs hidden; label-version agreement with the earlier references ranges from 85.3% to 91.3%. Under the reviewed labels, ProEviMem has higher IS than CSM in all three scopes; the small AS ordering changes between label versions (Appendix D). The provisional paired episode diagnostic distinguishes first assistance, valuable follow-up, and repetition, characterizing coverage and selectivity under a shared opportunity inventory (Appendix D.1).

6 Related Work

Agent memory and long-horizon retrieval. Generative Agents, MemoryBank, and MemGPT organize observations and experience through reflection, updating, or tiered context management (Park et al., 2023; Zhong et al., 2024; Packer et al., 2023). Mem0, A-MEM, and HippoRAG develop extracted memories, linked records, and graph-based retrieval (Chhikara et al., 2025; Xu et al., 2025; Gutiérrez et al., 2024). Graphiti maintains event-derived relations with temporal validity and provenance for retrieval (Zep, 2026). ProEviMem derives recurring conditional patterns from daily trajectories, assesses their historical support, and matches applicable patterns to incoming events.

Proactive decision and memory. ProactiveAgent studies whether an agent should propose a task or remain silent and uses annotated event data to elicit this capability (Lu et al., 2025). ContextAgent combines sensory contexts with historical personas to anticipate service needs (Yang et al., 2025), while PASK couples streaming demand detection with hybrid long-term memory (Xie et al., 2026). Memory-oriented proactivity also supports internal reasoning and execution: MemCog integrates proactive memory exploration into conversational reasoning (Li et al., 2026), and Remember When It Matters uses a separate memory agent to inject selected reminders into long-horizon action agents (Wu et al., 2026). ProEviMem focuses on constructing service-relevant temporal memory through source checking, historical replay, and current-event matching.

Evaluation of proactive service. LoCoMo and LongMemEval assess long-term conversational memory and reasoning (Maharana et al., 2024; Wu et al., 2025). Proactive benchmarks examine task proposals, latent-need detection, or memory triggering (Lu et al., 2025; Xie et al., 2026; Li et al., 2026). ProactiveStream evaluates sequential INTERVENE/ABSTAIN decisions as agents accumulate memory within isolated, causally replayed event streams, using episode-aware references to assess intervention timing. Research on notification timing further shows that interruption cost depends on task phase and receptivity (Adamczyk and Bailey, 2004; Iqbal and Bailey, 2007, 2008; Mehrotra et al., 2016). This motivates evaluating opportunity

recovery together with unnecessary intervention.

7 Conclusion

We presented ProEviMem, a proactive memory framework that combines bounded recent events, source-checked daily trajectories, and replay-assessed temporal patterns. Across the three scopes of ProactiveStream, ProEviMem achieves higher Intervention Score and Balanced Accuracy than the evaluated baselines and cross-week summary control. Ablations support the contributions of hierarchical organization, daily fact validation, and historical replay gating. These findings highlight the value of matching source-grounded, historically supported memory to current observations for service-opportunity detection.

Limitations

ProactiveStream covers three low-risk service scopes and 17 streams under a protocol fixed after joint method and benchmark development. Independent held-out evaluation and cross-family model transfer remain future work. Reference labels represent reviewed opportunities; the 450-target review assesses three selected streams, while episode judgments remain partially confirmed. The RA suppression diagnostic holds later states fixed; controller-free behavior requires a sequential comparison. Historical replay assesses observed support; prospective validity under changing behavior requires further study. Our offline metrics evaluate opportunity detection. Recipient benefit, interruption burden, end-to-end cost, and deployment safety require prospective measurements. The reported results support comparisons within the evaluated protocols, with broader generalization depending on additional populations, observation conditions, and model families.

Ethical Considerations

Our evaluation is limited to optional activity support, gentle check-ins, and low-risk repository assistance. Medical and high-risk safety decisions require separate domain-specific validation and safeguards. Inference is isolated by stream, and reference labels, episode annotations, and scoring metadata are excluded from model inputs.

Longitudinal events and derived memories can reveal sensitive routines even when identifiers are pseudonymized. Event content, prediction traces,

and inferred trajectories require access and retention protections; external-model use should minimize transmitted information and respect source-data permissions.

Deployment should require informed opt-in and allow inspection, correction, deletion, retention control, and disabling proactive assistance. An intervention prediction represents a service proposal; consequential action requires separate authorization. These deployment safeguards require separate implementation and evaluation.

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A Pipeline-Level Model Sensitivity

Table 4 provides the complete model-sensitivity comparison summarized in Section 5.4. The decision LLM and the LLMs constructing daily trajectories and temporal patterns are replaced together. The benchmark, memory architecture, role-specific prompts, decoding configuration, validation thresholds, and controller remain fixed; the resulting memory content may change. ProEviMem shows increasing IS and decreasing False Positive Rate from Luna to Terra to Sol in every scope. This comparison measures the joint effect of changing decision and construction models. All hierarchy and validation ablations retain Luna.

Table 4: ProEviMem model sensitivity (%). Decision and construction LLMs change together. Bold marks the best value per scope.

Scope	Model	IS $\uparrow$	FPR $\downarrow$
AS	Luna	48.8	46.1
	Terra	54.1	39.4
	Sol	58.6	34.4
AL	Luna	58.6	10.3
	Terra	64.4	9.1
	Sol	69.3	7.5
RA	Luna	55.4	18.1
	Terra	60.3	14.9
	Sol	66.4	11.8

B Memory-Construction Audit and Online Availability

Complementing the evidence-check ablations in Section 5.3, we audit existing memory-construction artifacts and maintenance logs from Activity Support and Ambient Living, covering nine participants and four homes. Table 5 distinguishes validation and replay operations from unique retained artifacts. The audit checks source support, entity consistency, and replay consistency.

Retained evidence. The daily store contains 423 trajectories, with 367 from Activity Support and 56 from Ambient Living, comprising 1,279 retained segments. All retained segments pass the configured source-support and entity-consistency checks, with all evidence linked to the same entity. Separately, construction logs record 311 rejected daily validation checks. These observations characterize construction errors, their filtering, and the resulting retained evidence.

The final temporal-pattern store contains 54 source-linked proposals whose recorded replay results are consistent with offline recomputation: 42 are Active and 12 remain Candidate. Decision contexts use lifecycle-eligible patterns that satisfy current-applicability conditions (Appendices E.3 and E.4). Across weekly maintenance updates, 118 of 140 replay attempts pass the activation gate, while 22 remain below its thresholds. These operation counts are distinct from the 42 unique Active patterns in the final store.

Online availability. Daily Trajectory Memory first becomes available for all 13 entities at a median elapsed time of 0.80 days. An Active temporal pattern first becomes available for 11 entities at a median of 6.79 days. Current-applicability matching first supplies a temporal pattern for eight entities at a median of 10.46 days. Each median is calculated over entities reaching the corresponding stage. Pattern activation and current relevance are checked separately: admitted patterns enter a decision context after current-applicability matching.

These observations show that daily and temporal-pattern evidence can become available within the evaluated streams, leaving subsequent events for evidence-informed decisions. The update order in Section 3.3 preserves the causal boundary: evidence used to judge an event is constructed from previously committed events. The construction checks complement the RQ3 validation ablations in Section 5.3; the stage-specific availability statistics address RQ4.

Table 5: Memory-construction audit for Activity Support and Ambient Living (nine participants and four homes). Log rows count maintenance operations; store rows count unique retained artifacts.

View	Counting unit	Audit result
Daily validation log	validation attempt	311 rejected before storage
Retained daily store	unique evidence-linked segment	1,279 checked; all source-supported; 0 cross-entity
Weekly replay log	replay attempt	140 attempts; 118 pass; 22 remain below the gate at that update
Retained pattern store	unique temporal-pattern proposal	54 checked; 42 Active; 12 Candidate; 0 inconsistent replays

C Illustrative Decisions and Error Cases

Table 6 traces one recorded AS decision from the 450-target diagnostic run, selected because its source events, daily artifacts, contemporaneous weekly proposal, and matched-pattern telemetry can be linked. It uses the proposal version available at decision time. The example illustrates evidence flow through one recorded decision.

The current step counts provide contemporaneous activity evidence for assessing the earlier STILL pattern’s service relevance. We interpret the logged decision using the recorded current observations and matched pattern. The logged ABSTAIN originates from the decision LLM and passes through the controller unchanged, illustrating separate pattern-applicability and intervention judgments.

The separate frozen 16-case inventory includes two false positives and two false negatives from Event-only logs. In Activity Support, the false-positive case receives an activity suggestion following substantial movement during the preceding hour; the false-negative case receives ABSTAIN at a reference-positive low-activity event. In Ambient Living, the false-positive case receives a check-in during ordinary bedroom–bathroom movement; the false-negative case receives ABSTAIN during a reference-positive quiet period spanning two consecutive segments.

These errors illustrate the distinction between an observable condition and an actionable service opportunity. These cases motivate joint assessment of current activity, historical context, and actionable service needs. Historical evidence contextualizes these observations for the agent’s final service-value judgment. Repeated windows additionally require assessing whether evidence or service needs have changed, complementing the controller trade-off examined in Section 5.3.

This qualitative analysis addresses the failure-analysis part of RQ4 and illustrates why ProE-

viMem separates memory evidence, present service value, and prior opportunity coverage.

D Independent Human Review Across Three Scopes

We conduct a sampled label review within the main benchmark, selecting 150 consecutive target events per scope (450 in total). A separate three-reviewer team independently annotates these segments while blinded to ProEviMem and CSM outputs, and discusses its judgments to obtain consensus references. Each segment belongs to exactly one participant, home, or project repository. A fixed SHA256-based rule first ranks eligible streams and then eligible contiguous starts using public metadata only, and requires at least seven days of preceding history. The selected streams are AS participant_35, AL home_0004, and the RA repository callstack/react-native-pager-view. Their target periods are December 16, 2015–January 18, 2016; June 23–28, 2011; and June 21–November 4, 2025, respectively. The corresponding ProEviMem pipeline processes all 37, 203, and 152 preceding events chronologically to provide historical context; scoring covers the 150 target events in each scope. Thus, 842 public events supply causal context for 450 scored targets; each target can use only its current observation and strictly prior same-stream history.

The reviewers inspect each target together with its preceding same-stream history. The confirmed review yields 27/123, 15/135, and 34/116 INTERVENE/ABSTAIN labels for AS, AL, and RA, respectively. Agreement with the earlier diagnostic references is 88.7% (133/150), 91.3% (137/150), and 85.3% (128/150) for AS, AL, and RA, respectively. In AS and AL, 13 and 13 labels change from INTERVENE to ABSTAIN, compared with 4 and 0 changes in the reverse direction; RA has 6 removals and 16 additions. Most judgments agree, with consequential differences at service-

Table 6: A source-to-decision timeline for one AS participant. Dates are in December 2015; times use the stored UTC timestamps. Text condenses source events and construction/decision logs.

Stage and recorded time	Source-linked observation or operation
Historical events Dec. 15, 17–19, 21	Participant 35 has recorded STILL observations in slots 2–4 across five cited days. For example, Dec. 15 at 16:32:38 records slot 2, work, STILL, and 586 steps in the preceding hour (event 5fd72a80. . .). STILL describes activity at the recorded observation time.
Daily construction Dec. 16–22	Each cited day becomes a source-linked daily trajectory at the next daily rollover. The Dec. 15 trajectory retains the example’s event reference and observed fields; its recorded creation time is Dec. 16 at 12:02:49.
Weekly proposal Dec. 22	At the weekly update on Dec. 22, the proposal cites the five completed trajectories dated Dec. 15, 17–19, and 21: recognized activity repeatedly equals STILL in slots 2–4, with applicability window 16:00–22:00. The maintenance record at 12:02:27 reports one proposed and one activated pattern.
Current event Dec. 24, 16:31:26	Event c956aa51. . . : slot 2, available, in-transit status false, store location, activity UNKNOWN, 256 steps in the preceding 30 minutes and 778 in the preceding hour. The context contains 12 prior events, two daily trajectories, and the matched pattern 93463c34. . . .
Decision, then commit	Both model proposal and final output are ABSTAIN; controller action is ALLOW, preserving the model output. Telemetry records that the current event is committed after context construction. Here, the matched recurring observation accompanies an ABSTAIN decision.

opportunity boundaries. The reported counts quantify agreement between label versions and the directions of label changes. These annotations serve as references for the additional evaluation; the main benchmark labels and result tables remain unchanged. For model comparison, we use the recorded ProEviMem and CSM decisions from the selected-segment diagnostic runs, which are separate from the main runs. For each method, exactly the same target decisions are scored against both label versions; rescoring preserves the recorded outputs. Like the main experiment, this diagnostic uses gpt-5.6-luna through the official OpenAI API, with temperature zero and thinking disabled. Decision-output limits are 256 tokens for AS/AL and 512 for RA. Both methods retain the 72-hour/100-event bound and episode control; ProEviMem additionally retains daily validation and replay gating.

Table 8 reports ProEviMem under the confirmed event-level references, and Table 7 adds CSM under both label versions. ProEviMem identifies 24/27 AS, 15/15 AL, and 18/34 RA opportunities. Its IS remains above CSM in AL and RA under both versions; the small AS difference changes direction. Thus, the paired audit reveals both agreement and label-sensitive method rankings. For example, AL ProEviMem IS changes from 68.7% to 55.6% when only the references change, while CSM changes from 46.4% to 37.2%. These comparisons hold the selected segment and its diagnostic outputs fixed; the main evaluation uses its own full-scope events and runs. Across scopes, im-

mediate and scheduled service decisions share the INTERVENE label. The paired episode analysis in Appendix D.1 uses references grounded in these reviewed events and their causal histories, with its annotation status specified below.

Table 7: Paired reference-label sensitivity on 150 targets per scope. IS is in percent. Within each method, the recorded diagnostic-run decisions are identical in both columns; only the reference version changes. Earlier references are original labels for AS/AL; RA uses 122 original labels and 28 subsequently completed human labels; the RA reference version therefore combines original and subsequently completed annotations.

Scope	Method	Earlier IS	Reviewed IS
AS	CSM	44.4	41.0
	ProEviMem	42.6	42.5
AL	CSM	46.4	37.2
	ProEviMem	68.7	55.6
RA	CSM	49.4	46.5
	ProEviMem	57.6	47.4

D.1 Paired Episode-Level Behavioral Diagnosis

On the same 450 targets and 392 preceding events, we compare recorded CSM and ProEviMem decisions using a common set of 67 proposed episodes: 26 in AS, 10 in AL, and 31 in RA. The analysis separates first effective assistance, useful follow-up, and repetition, using fixed recorded outputs and event labels. Each episode specifies a service goal and object, causal evidence, onset, valid observed event positions, and any explicit deadline or closure evidence. This post-hoc paired diag-

Table 8: Recorded ProEviMem decisions evaluated against consensus labels from a team blinded to model outputs, on 450 targets with 150 contiguous decisions per scope. Each segment and its prefix are entity-isolated. Counts use final intervention decisions after episode control, including immediate and scheduled outputs. Prefix events supply historical context.

Quantity	AS	AL	RA
Scored decisions	150	150	150
Context-only preceding events	37	203	152
Reference interventions	27	15	34
Reference abstentions	123	135	116
ProEviMem interventions	86	39	42
TP	24	15	18
FP	62	24	24
FN	3	0	16
TN	61	111	92
Intervention Rate	57.3%	26.0%	28.0%
Precision	27.9%	38.5%	42.9%
ODR	88.9%	100.0%	52.9%
IS	42.5%	55.6%	47.4%
False Positive Rate	50.4%	17.8%	20.7%
Balanced Accuracy	69.2%	91.1%	66.1%

nostic uses model-assisted episode references and semantic matches, assessed separately from the blinded event-label review. Six AL follow-up classifications have human confirmation; the remaining episode and matching proposals retain provisional status. Validity is assessed at the listed observations in each episode reference.

Reference and model histories. Fixed event labels use reference-opportunity novelty (Equation 9); episode coverage follows the evaluated method’s actual preceding interventions. Each method initializes coverage from its own prefix outputs. Scoring counts only target interventions, and each method maintains an independent coverage history. A target intervention must match an episode’s service goal and object at a listed valid observation. For timeliness, intended service time is the decision time plus the specified delay, with zero delay for immediate intervention. If an explicit reference deadline exists, coverage credit requires the intended service time to meet that deadline. The first timely match covers the episode. For subsequent matches, we compare the message and available evidence with that method’s preceding assistance for the same episode. A follow-up is valuable when new causal evidence changes the actionable service or next step and the message addresses that change; rephrasing the same advice or continued status alone is redundant. Cases re-

quiring unobserved delivery, recipient response, or task completion are left uncertain. Valuable follow-ups contribute to service quality within the existing episode; coverage counts each episode once. Each output credits at most one episode: among compatible references, the first uncovered one in the fixed reference order takes precedence; otherwise the first already covered one is used.

Metrics. Let I be all target interventions, F first valid interventions, V valuable follow-ups, and R redundant follow-ups. Episode Coverage is covered new episodes divided by all new episodes. Supported Intervention Precision (SIP) is $(F + V)/I$, extending the first-only Episode Precision F/I while retaining single-credit episode coverage. Redundant Intervention Rate (RIR) is R/I . All target interventions contribute to I in SIP and RIR, including uncertain follow-ups (U), unmatched outputs, late outputs, and unresolved matches. Their numerators count supported assistance ($F + V$) and redundant follow-ups (R), respectively; Episode Coverage uses the number of new reference episodes as its denominator. These categories describe compatibility with the proposed references. Recipient benefit requires prospective validation. Table 9 reports coverage and rates; Table 10 retains the count decomposition.

Observation boundaries. The main diagnostic treats the first AL opportunity as target-onset; a plausible continuation from the prefix changes both methods’ denominator to nine, with CSM covering 7/9 and ProEviMem 9/9. Coverage describes the observed segment; lifetime coverage requires follow-up beyond that segment. RA has six late CSM outputs and four late ProEviMem outputs; scheduled decisions are assessed using their intended service times and explicit deadlines. Three CSM RA follow-ups have uncertain value: accepting all three would raise its SIP from 33.8% to 38.5%, a sensitivity scenario for those three judgments. Alternative service interpretations of other unmatched and unresolved outputs remain open for review.

Behavioral comparison. In AS, both methods achieve the same coverage, with fewer interventions from ProEviMem. In AL, ProEviMem covers two more proposed opportunities and provides more valuable follow-ups. Its redundant-intervention rate is also higher. In RA, CSM covers more opportunities. ProEviMem achieves higher

Table 9: Paired episode diagnostic on 150 targets per scope with partially confirmed references and matches. Both methods use the same references and their own causal histories. Rates are percentages. SIP credits first valid assistance and valuable follow-up; RIR counts redundant follow-up. Bold marks the better unrounded directional value within each scope (or both for ties).

Scope	Method	Covered / New	Coverage $\uparrow$	SIP $\uparrow$	RIR $\downarrow$
AS	CSM	23/26	88.5	26.7	1.1
	ProEviMem	23/26	88.5	27.9	1.2
AL	CSM	8/10	80.0	39.3	32.1
	ProEviMem	10/10	100.0	41.0	43.6
RA	CSM	18/31	58.1	33.8	4.6
	ProEviMem	14/31	45.2	40.5	9.5

Table 10: Intervention counts underlying Table 9. I equals the sum of all seven outcome categories. U denotes uncertain follow-up value; Unresolved denotes a pending semantic or condition match. Both remain separate from redundant or supported assistance. Each episode contributes once to the episode count.

Scope	Method	I	F	V	R	U	Unmatched	Late	Unresolved
AS	CSM	90	23	1	1	0	64	0	1
	ProEviMem	86	23	1	1	0	60	0	1
AL	CSM	28	8	3	9	0	8	0	0
	ProEviMem	39	10	6	17	0	6	0	0
RA	CSM	65	18	4	3	3	30	6	1
	ProEviMem	42	14	3	4	0	16	4	1

SIP with fewer interventions. The supported shares measure alignment with this finite reference inventory; the value of remaining messages requires additional semantic review. These trade-offs characterize coverage, supported follow-up, and repetition under the shared diagnostic rubric. Controller-specific causal effects require a controlled sequential comparison.

E Memory Implementation Details

This appendix expands the evidence lifecycle in Section 3; the core acceptance conditions and causal interfaces remain in the main paper.

E.1 Daily Buffer and Checkable Fields

Daily construction uses an independent buffer that preserves all committed events of the open local day alongside the bounded Event Memory. The buffer is cleared after that day is processed. Each daily segment records a local interval, typed observed, context-before, and outcome fields, and supporting event identifiers. These activity segments are distinct from service episodes used for intervention control. Scalar values are canonicalized; field paths must agree exactly or through a schema-preserving suffix. The validator also checks entity, interval, and citation consistency. Accepted facts

have matching source values and resolved same-entity citations. Validation and replay operate on typed evidence; free text supplies accompanying descriptions. Accepted segments form daily trajectories retained in the current construction cycle, even after their source events leave Event Memory.

E.2 Construction Clock and Replay Boundaries

A weekly cycle is a time-ordered block of seven completed daily-trajectory records for one entity; each record belongs to one block. Cycles advance with available daily records, so a cycle may span more than seven calendar days. Empty or unobservable records advance cycle completion while leaving fulfilled and violated replay counts unchanged. After consolidation, the online daily-memory view starts a new cycle, while daily records remain available for audit.

A proposal specifies trigger context g , expected pattern p , a local-time window, optional weekday restrictions, and supporting daily identifiers. It describes a conditional behavioral expectation. Replay evaluates days satisfying the applicability conditions and assigns censored status to missing observations. An observable day with a matching trigger is fulfilled if p occurs at or after the trigger

within the specified window; otherwise it is violated. Each date contributes at most once. The confidence denominator in Equation 6 includes only fulfilled and violated days. Confidence is zero when the eligible-day count is zero; lift is zero when the base rate is zero. The base rate uses observable days under the same window and weekday restrictions, including both triggered and other observable days.

E.3 Pattern Lifecycle

Lifecycle status controls retrieval eligibility, separately from current occurrence and service-episode states. After admission, fulfilled and violated observations update stored confidence through a smoothed support ratio combining replay counts and subsequent outcomes with one initial success and one initial failure. Censored observations leave success and failure counts unchanged. Two consecutive violations move an Active pattern to *weakening*, which remains retrievable at lower priority; four move it to *dormant*. A fulfilled observation restores a weakening pattern to Active when updated confidence is at least 0.5. A dormant pattern returns only to Candidate and requires replay readmission. The retrieval-eligible lifecycle states are Active and weakening.

E.4 Current Occurrence Matching

A deterministic matcher checks trigger context, weekday restrictions, and expected windows using a 30-minute lead and grace interval. Triggers may be current or previously recorded causal occurrences. Each occurrence has one of five states: upcoming, pending, fulfilled, deviated, or censored. Retrieval considers upcoming, pending, and deviated occurrences of lifecycle-eligible patterns. These occurrences are ranked by state (deviated, pending, then upcoming), lifecycle priority, stored confidence, and temporal distance; at most three patterns enter R_t . All matching and state assessment use only the current event and causally available observations. Daily Trajectory Memory supplies all completed daily trajectories in the current cycle.

E.5 Episode Identity and Causal Execution

In AS/AL, the controller forms an intervention key from scope-specific episode state, proposed service action and target, and sorted pattern identifiers present in the decision context. The identifier component contains the explicit pattern identi-

fiers available in context and defaults to an empty list for plain summaries or graph facts. State includes availability/activity context for AS and activity/room context for AL. AL additionally distinguishes onset, confirmed, and sustained phases at one, two, and three or more consecutive matching observations, respectively. When an explicit episode key is unavailable, scope-specific time or target fields identify the opportunity, and unchanged evidence is additionally required for duplicate suppression. Changed episode state, a restarted state sequence, changed service or pattern identifiers, or a new local day can produce a different intervention key; this permits the decision model to reassess service value.

In RA, matching uses repository-object identity and the most recent allowed intervention’s mode, recipient, content, and firing condition. An exact match suppresses a repeated create proposal by converting it to ABSTAIN. Changed signatures or other episode operations pass this duplicate check. Re-entry follows these identity and phase rules, with each intervention originating from an LLM proposal. All main-table methods and memory-level variants share this policy; each method’s available evidence determines its proposal’s identity.

The causal execution order is:

1. Prepare prior memory; any due maintenance uses only previously committed events.
2. Stage e_t as the current input, defer its memory commit until the decision is finalized, and select daily and temporal evidence.
3. Obtain the decision LLM’s proposal from the structured context.
4. Apply repeated-opportunity suppression and finalize the output.
5. Commit e_t and update coverage state; maintenance incorporating e_t affects only later decisions.

Reference labels are joined only for offline scoring.

F Public Events and Reference-Label Protocol

F.1 Scopes and Public Events

Activity Support. This scope derives from HeartSteps mobile activity contexts (Klasnja et al., 2019). We retain 1,456 meta-events from nine complete participant streams, covering 367 observed

participant-days. Complete participant streams preserve the chronology required for daily and longer-term memory construction. Public fields include decision-time availability, recognized activity, recent step summaries, location category, weather, and local decision slot. Randomized treatment assignments and future behavioral outcomes are excluded. Reference labels concern opportunities for optional activity support.

Ambient Living. This scope derives from CASAS ambient-sensor traces (Cook et al., 2013) and contains four homes with 14 consecutive days each. Within every local day, ten-minute preprocessing blocks are merged into 28 regular 50-minute segments and one 40-minute tail, producing 406 events per home. Every source block is assigned to exactly one segment, preserving the complete two-week chronology. A segment becomes available only after its interval closes and contains within-interval motion counts, room sequences, door actions, and compact numeric summaries. Each public segment describes its own interval; the evaluated agent derives historical evidence from preceding same-home observations. The service space is restricted to gentle, low-risk check-ins and excludes disease, fall, travel, danger, and unsupported preference inference.

Repository Assistance. This scope organizes 1,459 repository activity records into four isolated streams, with 174 reference interventions (Table 1). Each event represents a timestamped repository activity, such as a commit push, issue update, pull-request action, or review. We preserve GitHub’s native event granularity, with each event corresponding to one intervention decision. Within each repository, records are ordered by event time, with event ID used to break ties deterministically. The source inventory includes public GH Archive records (Grigorik, 2026). Each repository independently maintains Event Memory, Daily Trajectory Memory, and Temporal Pattern Memory, together with its intervention history and episode state. Observations, extracted evidence, and coverage state remain local to each repository, including projects with shared contributors. The repository stream is the memory owner and evaluation unit. Services include low-risk triage, review, and maintenance assistance. Native PushEvent denotes repository activity; INTERVENE denotes an agent’s service decision.

Public-input contract. Reference-positive and reference-negative events follow the same scope-specific public schema. Public inputs contain only event identifiers, stream identifiers, timestamps, and contemporaneously observable fields. Reference labels, candidate flags, episode annotations, memory-level markers, and scoring masks remain evaluator-private; agents construct trajectory summaries and memory representations from public observations. The inventory includes early events observed before Daily Trajectory Memory or Temporal Pattern Memory becomes available.

F.2 Reference Labels and Causal Replay

Service-opportunity semantics. Annotators assess the current event e_t and the complete strictly prior public history $\mathcal{H}_{<t} = (e_1, \dots, e_{t-1})$ of the same entity, independently of the memory representation retained by a tested method. Let $A(e_t)$ indicate that the current event itself establishes a concrete, timely, feasible, and low-risk service opportunity, and let $B(e_t, \mathcal{H}_{<t})$ indicate that prior evidence makes such an opportunity visible at the current event. Let N_t^{ref} indicate opportunity novelty under the reference episode protocol below. The reviewed label follows Equation 9. These predicates specify annotation semantics. Annotators assess concrete service value in both routine and unusual situations, using current observations and prior evidence. Within a semantic episode, the first reliable and actionable window is labeled INTERVENE. Subsequent covered windows are labeled ABSTAIN unless the evidence strengthens, the feasible service changes, or the episode reopens. Reference episode coverage, and hence N_t^{ref} , is determined from the reviewed opportunity sequence up to the current event, independently of a method’s predicted interventions. The labels represent reviewed service opportunities.

Annotation protocol. An LLM-assisted screening stage identifies potential service opportunities and expands the review queue. Three team members then independently assign initial labels while blinded to the evaluated systems’ decisions, followed by team cross-review and adjudication. Every public event enters final review, covering the full inventory beyond the screening shortlist. The screening output prioritizes inspection; team adjudication determines each reference label. We use protocol-fixed evaluation on the jointly developed benchmark; independent held-out testing re-

mains future work. The Ambient Living aggregation rule is specified before its event rows are constructed, and each evaluation version jointly records its event inventory, reference-label map, intervention prompt, Event Memory boundary, and validation settings. For the main comparisons, these labels, prompts, memory bounds, and validation settings are frozen before the corresponding runs.

Causal replay. Every participant, home, or repository stream is replayed independently from empty method state and in timestamp order. At event t , a method may access the current event e_t , its own strictly prior same-stream memory, and any preceding outputs retained by its architecture. Access is restricted to the current public event and causally available same-stream state. The intervention decision is finalized before e_t is committed to memory. Daily and weekly maintenance involving e_t can therefore affect only later decisions. Reference labels remain evaluator-private during inference and are joined afterward for scoring. Annotation artifacts, reference episode states, and scoring meta-data remain organizer-private.

G Additional Experimental Configuration

External baselines. *ProactiveAgent* adapts ProactiveAgent’s released evaluation prompt and proposal/null output contract (Lu et al., 2025; THUNLP, 2026), using the study backbone. It retains causal event–decision history, capped at 1,000 events including the current event per request, as its memory interface. *Mem0* uses the open-source mem0ai==2.0.12 extraction, consolidation, and retrieval pipeline (Chhikara et al., 2025; Mem0, 2026), local all-MiniLM-L6-v2 embeddings, top-8 same-entity retrieval, and a 0.1 search threshold. The baselines retain their method-specific memory and prompt interfaces while sharing the task semantics and final Episode Controller. Each method retains its configured context and token budgets.

A-MEM uses linked memory notes and association updates (Xu et al., 2025); retrieved notes augment the bounded event context. *ContextAgent-ICL* uses the in-context proactive-service formulation of ContextAgent (Yang et al., 2025), adapted to the public event representation and study backbone. These event-stream adaptations are evaluated on the same scope-specific reference labels as the

other main-table methods.

Cross-week summary control. CSM is an in-house plain-summary control. It summarizes each completed local day and, after seven completed daily records, combines that block with the previous cross-week summary. This recursive update combines retained information from earlier weeks with the latest block. The decision context contains the current event, bounded Event Memory, completed daily summaries in the current block, and the latest cross-week summary. The adapter caps daily and cross-week summaries at 2,000 and 6,000 characters, respectively. CSM checks output format and length before supplying the summary text directly to the decision model.

Temporal-graph baseline. Graphiti uses the open-source temporal-graph framework (Zep, 2026) to ingest committed public events into a stream-isolated graph. The adapter uses Graphiti 0.30.2, Neo4j, and local 384-dimensional all-MiniLM-L6-v2 embeddings. The current public event is the retrieval query; native hybrid edge search with reciprocal-rank fusion returns at most ten relation facts, including validity fields and source-record identifiers. The adapter supplies the retrieved relation facts and their source identifiers directly. Graphiti’s ingestion episodes denote source records; evaluation episodes denote service opportunities. Both baselines retain ProEviMem’s 72-hour/100-event bound for the event context and reuse the task policy, output contract, and Episode Controller with memory-specific guidance. CSM supplies summary text and Graphiti supplies relation facts through their respective memory interfaces. The comparison evaluates alternative memory interfaces with method-specific prompt guidance and computational budgets.

Memory variants and retrieval control. *Event-only* receives the current event and bounded Event Memory (Equation 4). *Event+Daily* additionally receives fact-validated daily trajectories; ProEviMem also receives eligible temporal patterns (Equation 7). These three variants share the intervention prompt, output schema, and Episode Controller, differing in available memory evidence. For Activity Support and Ambient Living, *Flat Retrieval Memory* adds to Event-only the three most similar older same-entity events outside its window. Its retrieval path consists of fixed dense embeddings, cosine-similarity ranking, and direct inclu-

sion of the selected source events.

Mechanism ablations and controller diagnostic. Section 5.3 defines two evidence-check ablations and a suppression diagnostic; Table 12 reports their intervention rates, precision, ODR, and IS. Each validation ablation removes only its named check. The controller diagnostic compares recorded proposals with finalized outputs on the same RA sequence, holding later states and decisions fixed.

Model roles and implementation. In the primary setting, decision and memory-construction calls use gpt-5.6-luna through the official OpenAI API, with temperature zero and thinking disabled. Decision generation and memory construction have separate call roles and output budgets. Decision limits are interface-specific (256 or 512 tokens); construction-call limits are configured separately and recorded by role in the run configuration. Event Memory contains at most 99 prior events within 72 hours, plus the separately supplied current event. Daily construction follows local-day closure; weekly construction and replay follow seven completed daily records as defined in Section 3. All memory and validation records are entity-scoped.

We also test gpt-5.6-terra and gpt-5.6-sol. Both decision and construction LLMs change together; Appendix A provides the fixed settings and complete results. Input manifests, prompts, schemas, thresholds, and role-specific configurations are versioned in the experiment ledger.

Scoring. Section 5.1 defines the scoring policy and metrics. Using event-level confusion counts, precision is $TP/(TP + FP)$, Opportunity Detection Rate is $ODR = TP/(TP + FN)$, and Intervention Score is $IS = 2TP/(2TP + FP + FN)$. IS is the harmonic mean of precision and ODR; the reported percentages use the same reference labels and finalized decisions as the other event-level metrics. Intervention Rate measures decision frequency.

G.1 Intervention-Rate Interpretation

Intervention Rate describes a behavioral operating point interpreted alongside coverage and selectivity. Writing ρ for reference-positive prevalence, the exact identity $IR = \rho ODR + (1 - \rho) FPR$ separates opportunity recovery from false intervention, with all rates expressed as fractions. The prevalences are 25.1%, 11.7%, and 11.9% in AS,

AL, and RA. Scope-specific decision behavior also contributes to the variation: ProactiveAgent’s intervention rates are 9.2% on AL and 53.1% on RA, whose reference-positive prevalences are similar. Their respective ODR and false-intervention rates reveal different coverage–selectivity operating points under the scope-specific evidence and service semantics. This descriptive decomposition summarizes the joint behavior of the LLM, memory, prompt, and controller.

H Extended Ablation and Controller Results

Tables 11 and 12 expand Figure 3 and Table 3 with net TP increments, intervention rates, and ODR. Net TP changes summarize differences between aggregate method counts. Intervention rates and ODR are retained to expose coverage–selectivity trade-offs, including the RA-only Episode Controller comparison.

Table 11: Memory-hierarchy ablation. IS and False Positive Rate (FPR) are percentages; Δ TP is a net count. Daily Δ TP compares Event+Daily with Event-only; Pattern Δ TP compares ProEviMem with Event+Daily. The increments summarize aggregate count differences. Bold marks higher IS and lower FPR. Count increments are reported per scope.

Scope	Method	IS $\uparrow$	FPR $\downarrow$	Daily Δ TP	Pattern Δ TP
AS	Event-only	38.3	51.3	–	–
	Event+Daily	38.9	51.3	+4	–
	ProEviMem	48.8	46.1	–	+57
AL	Event-only	40.0	12.4	–	–
	Event+Daily	45.1	12.4	+15	–
	ProEviMem	58.6	10.3	–	+33
RA	Event-only	28.7	18.9	–	–
	Event+Daily	48.5	19.1	+64	–
	ProEviMem	55.4	18.1	–	+22

Table 12: Evidence-check ablations and controller diagnostic (%). Validation rows remove the named check; Before suppression scores proposals from the recorded RA run while holding later states fixed. ProEviMem is last. Bold marks the best directional score. The controller diagnostic is reported for RA alone.

Scope	Method	Intervention	Precision $\uparrow$	ODR $\uparrow$	IS $\uparrow$
		Rate			
AS	w/o Replay Gate	55.0	31.1	68.2	42.7
	w/o Daily Validation	55.8	34.4	76.7	47.5
	ProEviMem	53.8	35.8	76.7	48.8
AL	w/o Replay Gate	19.3	43.6	72.1	54.4
	w/o Daily Validation	18.9	45.9	74.2	56.7
	ProEviMem	17.7	48.6	73.7	58.6
RA	w/o Replay Gate	29.4	33.8	83.3	48.1
	w/o Daily Validation	29.6	36.1	89.7	51.5
	Before suppression	32.4	36.7	99.4	53.6
	ProEviMem	26.7	40.1	89.7	55.4